Economics S5040: Dynamic Stochastic Modeling
Lecture 6 — Finite Markov Chains

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Fall 2026

1 From Random Transitions to Markov Chains

In Lecture 5 we considered a population divided between employment and unemployment. Suppose that the probability of tomorrow's labor-market status depends only on today's status.

Let

$$P = \begin{pmatrix} 0.90 & 0.10 \\ 0.40 & 0.60 \end{pmatrix}.$$

The rows describe today's state and the columns describe tomorrow's state. Thus

$$P_{ij} = \Pr(X_{t+1} = j \mid X_t = i).$$

If state 1 is employment and state 2 is unemployment, then an employed worker remains employed with probability 0.90, while an unemployed worker finds a job with probability 0.40.

Each row sums to one:

$$\sum_j P_{ij} = 1.$$

A matrix with nonnegative entries whose rows sum to one is called a **stochastic matrix** or **transition matrix**.

A stochastic process $\{X_t\}$ is a **Markov chain** if, conditional on the current state, the past provides no additional information about the next state:

$$\Pr(X_{t+1} = j \mid X_t = i, X_{t-1}, \ldots, X_0) = \Pr(X_{t+1} = j \mid X_t = i).$$

This is the **Markov property**. It does not say that history is irrelevant. History can matter because it determines the current state. It says that once we know the current state, we do not need the rest of the history to forecast the next state.

2 The Distribution of States

Suppose that today 80 percent of workers are employed and 20 percent are unemployed. Write the distribution as the row vector

$$\psi_0 = \begin{pmatrix} 0.8 & 0.2 \end{pmatrix}.$$

Tomorrow's distribution is

$$\psi_1 = \psi_0 P.$$

Using the matrix multiplication rule from Lecture 5,

$$\psi_1 = \begin{pmatrix} 0.8 & 0.2 \end{pmatrix} \begin{pmatrix} 0.90 & 0.10 \\ 0.40 & 0.60 \end{pmatrix} = \begin{pmatrix} 0.80 & 0.20 \end{pmatrix}.$$

In this particular example the distribution has not changed.

More generally,

$$\boxed{\psi_{t+1} = \psi_t P.}$$

This equation is one of the central equations of the lecture. It tells us how a distribution over states evolves over time.

In Python:

```
import numpy as np

P = np.array([[0.90, 0.10],
              [0.40, 0.60]])

psi = np.array([0.80, 0.20])

psi_next = psi @ P
print(psi_next)
```

The symbol `@` denotes matrix multiplication.

3 Two-Step and Multi-Step Transitions

Suppose we know a worker's state today and want the probability distribution of the worker's state two periods from now.

The answer is obtained from

$$P^2 = PP.$$

For example, the probability that an employed worker is unemployed two periods from now is

$$(P^2)_{12}.$$

Why? There are two possible intermediate states:

$$\Pr(E_t \to U_{t+2}) = \Pr(E \to E)\Pr(E \to U) + \Pr(E \to U)\Pr(U \to U).$$

Thus

$$(P^2)_{12} = (0.90)(0.10) + (0.10)(0.60) = 0.15.$$

Matrix multiplication is doing the probability accounting for us.

More generally,

$$\boxed{(P^n)_{ij} = \Pr(X_{t+n} = j \mid X_t = i).}$$

If the initial distribution is ψ_0, then

$$\boxed{\psi_t = \psi_0 P^t.}$$

In Python:

```
P2 = P @ P
P10 = np.linalg.matrix_power(P, 10)

psi_10 = psi @ P10
```

Notice the distinction between `P * P`, which multiplies entries element by element, and `P @ P`, which performs matrix multiplication.

4 Stationary Distributions

A distribution ψ^* is called **stationary** if applying the transition matrix leaves it unchanged:

$$\boxed{\psi^* = \psi^* P.}$$

For our employment example, write

$$\psi^* = \begin{pmatrix} e & u \end{pmatrix}, \qquad e + u = 1.$$

Stationarity requires

$$u = 0.10e + 0.60u.$$

Using $e = 1 - u$,

$$u = 0.10(1 - u) + 0.60u.$$

Hence

$$0.50u = 0.10, \qquad u = 0.20,$$

and therefore

$$e = 0.80.$$

So

$$\boxed{\psi^* = \begin{pmatrix} 0.80 & 0.20 \end{pmatrix}.}$$

This explains why our distribution in the previous section did not change: we happened to start at the stationary distribution.

The stationary distribution has a natural economic interpretation. If the transition probabilities remain fixed and the population distribution is stationary, then individual workers continue moving between employment and unemployment, but the *fraction* of workers in each state remains constant.

Stationarity of the aggregate distribution does not mean that individual agents stop moving between states.

5 Convergence Toward the Stationary Distribution

Now start away from the stationary distribution:

$$\psi_0 = \begin{pmatrix} 1 & 0 \end{pmatrix}.$$

We can repeatedly apply the transition matrix:

$$\psi_1 = \psi_0 P, \qquad \psi_2 = \psi_1 P, \qquad \dots$$

or equivalently

$$\psi_t = \psi_0 P^t.$$

The following Python code displays the first several distributions:

```
psi = np.array([1.0, 0.0])

for t in range(10):
    print(t, psi)
    psi = psi @ P
```

For this transition matrix, the distribution approaches

$$(0.8, 0.2).$$

We will not prove a general convergence theorem today. It is enough to understand that many economically useful Markov chains "forget" their initial distribution and converge toward a unique stationary distribution.

This is an important distinction:

- **stationarity** means that a distribution is unchanged by P;
- **convergence** asks whether distributions that begin elsewhere approach that stationary distribution.

6 Simulating an Individual History

So far we have followed probability distributions. We can also simulate the history of a single individual.

Suppose the worker is currently employed. We draw the next state according to the first row of P. If the worker is unemployed, we draw according to the second row.

```
rng = np.random.default_rng(1234)

P = np.array([[0.90, 0.10],
              [0.40, 0.60]])

state = 0 # 0 = employed, 1 = unemployed
history = [state]

for t in range(100):
    state = rng.choice([0, 1], p=P[state])
    history.append(state)
```

The simulated history is random. Running the simulation with another random seed will generally produce a different history.

But if we simulate for a long time, the fraction of periods spent in each state should be close to the stationary distribution.

For example:

```
history = np.array(history)

fraction_employed = np.mean(history == 0)
fraction_unemployed = np.mean(history == 1)

print(fraction_employed, fraction_unemployed)
```

This gives us a connection with the Law of Large Numbers and Monte Carlo methods from Lecture 5.

The stationary distribution describes long-run frequencies that we can compare with frequencies from a simulated sample path.

7 A Three-State Economic Example

Markov chains are not restricted to two states. Suppose aggregate productivity can be

$$z \in \{\text{low}, \text{normal}, \text{high}\}.$$

A possible transition matrix is

$$P = \begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.25 & 0.70 \end{pmatrix}.$$

For example,

$$P_{23} = 0.10$$

means that, conditional on productivity being normal today, there is a 10 percent probability that it will be high next period.

If

$$\psi_t = \begin{pmatrix} 0.2 & 0.6 & 0.2 \end{pmatrix},$$

then

$$\psi_{t+1} = \psi_t P.$$

Nothing conceptually changes when we move from two states to three, or from three states to twenty. This is one reason matrix notation is so useful.

In later lectures, the state of a dynamic economic problem will often evolve according to a Markov chain. The transition matrix will tell an optimizing agent how today's state affects the probabilities of tomorrow's possible states.

8 What to Remember

There are five ideas from today that we will use repeatedly:

1. A Markov chain has a state that evolves randomly over time.
2. The Markov property says that the current state contains the information needed to forecast the next state.
3. The transition matrix P contains the one-period transition probabilities.
4. Distributions evolve according to
$$\psi_{t+1} = \psi_t P, \qquad \psi_t = \psi_0 P^t.$$
5. A stationary distribution satisfies
$$\psi^* = \psi^* P.$$

We have also seen two complementary ways to study a Markov chain:

evolve the probability distribution	or	simulate individual histories.

Both approaches will be useful when we begin dynamic programming.